HDOC - 63rd day

FINDING THE MISSING NUMBER USING POINTERS

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array, nums[]. The array contains distinct integers from 0 to size, with exactly one number missing. Pass the array to a function named missing_number() using its address. Use pointers to pass the array between the calling function and missing_number(). The function missing_number() must:

- Find the missing integer from the range 0 to size.
- Display the missing number.
- Assume that all elements in the array are distinct.
- Use the fact that the array contains exactly n numbers from the range 0 to size, with one number missing.
- Display the result as an integer.

Algorithm

1. Start.

2. Read the size n of the array nums[].

3. Read n distinct elements into nums[]. Each element lies in the range 0 to n.

4. Pass nums and n to missing_number() using the array address.

5. Calculate the expected sum of all integers from 0 to n using n * (n + 1) / 2.

6. Initialize actual_sum = 0.

7. Using pointer arithmetic, add each array element *(nums + i) to actual_sum.

8. Calculate missing = expected_sum - actual_sum.

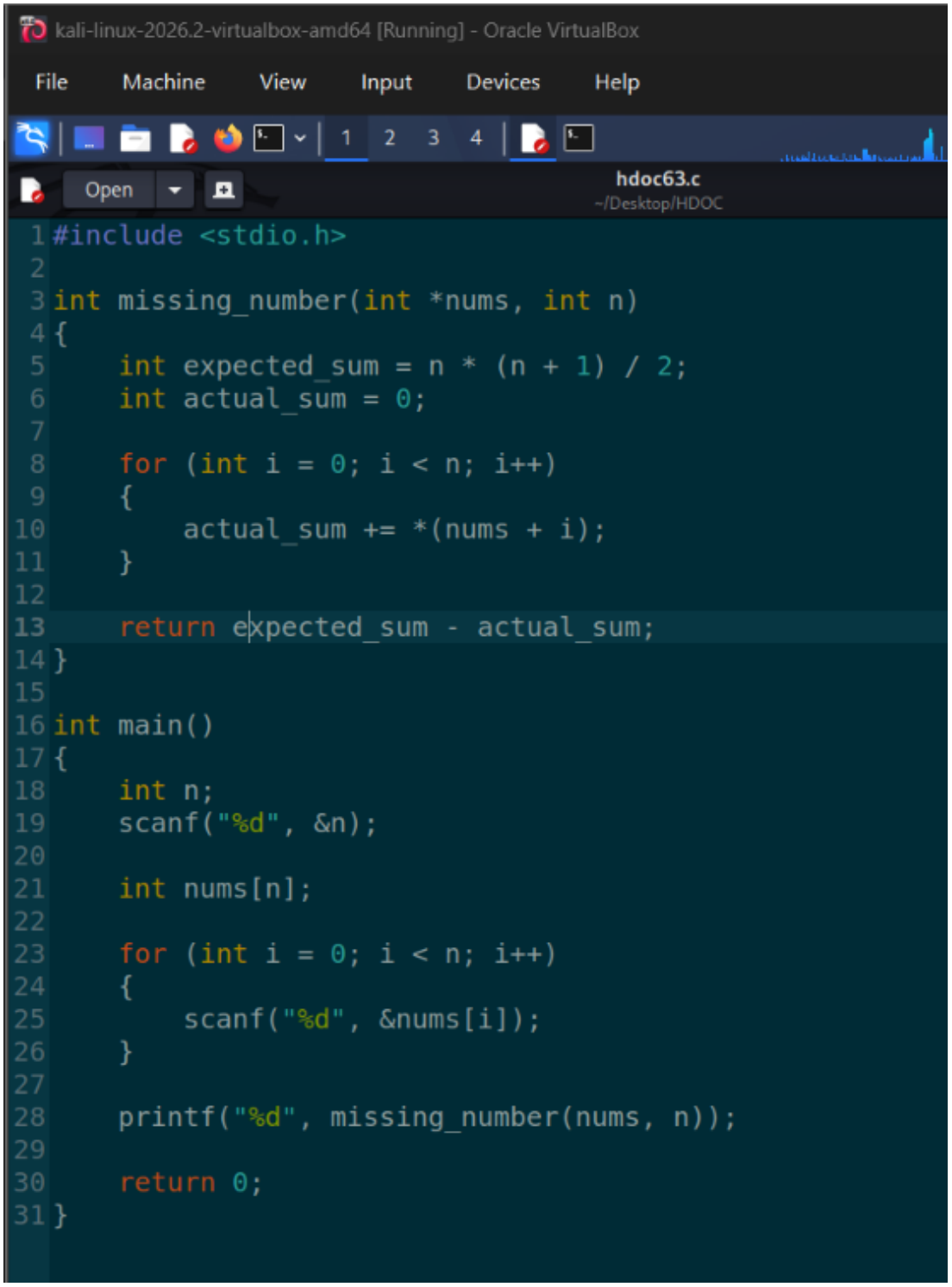
9. Display the missing number as an integer.

10. Stop.

Test Case Table

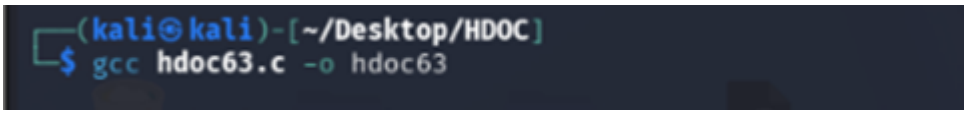
S. No.	Size	nums[]	Expected Output	Actual Output	Result
1	5	0 1 3 4 5	2	2	Pass
2	4	0 1 2 3	4	4	Pass
3	6	1 2 3 4 5 6	0	0	Pass
4	3	0 2 3	1	1	Pass
5	7	0 1 2 3 4 5 6	7	7	Pass

C Program


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File
Machine
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Open
hdoc63.c
~/Desktop/HDOC
#include <stdio.h>
int missing_number(int *nums, int n)
{
int expected_sum = n * (n + 1) / 2;
int actual_sum = 0;
for (int i = 0; i < n; i++)
{
actual_sum += *(nums + i);
}
return expected_sum - actual_sum;
}
int main()
{
int n;
scanf("%d", &n);
int nums[n];
for (int i = 0; i < n; i++)
{
scanf("%d", &nums[i]);
}
printf("%d", missing_number(nums, n));
return 0;
}

Compilation and Execution

Compilation command: gcc hdoc63.c -o hdoc63



Execution command: ./hdoc63

Execution Screenshot:

